

ShopCart Superstore Sales & Profit Analysis

Introduction

ShopCart is an e-commerce company that operates in North America. It sells products across different categories such as Furniture, Office Supplies, and Technology. Over the years, the company has experienced steady growth in sales and customer demand.

Despite this growth, the company began facing challenges related to profitability, operational efficiency, and sales performance across different product categories and regions.

To better understand these challenges, the company hired a Data Analyst to analyze historical sales data from 2022 to 2025 and identify the key factors affecting profitability., canyou make it more better

Business Objective

The primary objective of this project is to analyze ShopCart's historical sales data from 2022–2025 to evaluate sales performance and identify the key drivers of profitability across products, regions, categories, and customer segments in the United States.

Core Questions to Answer

1. **Business Summary:** What are the company's total sales, profit, and other key performance indicators (KPIs)?

2. **Trend Over Time:** How have sales and profits changed over time? Are there any noticeable seasonal patterns or trends?
 3. **Profitability Drivers:** Which products categories and sub-categories are the most & least profitable?
 4. **Regional Performance:** Which geographical regions and states contribute most to sales and profit? Are there any regions with high sales but low profitability ?
 5. **Customer Segment Analysis:** Which customer segment is most valuable ?
 6. **Discount Impact:** How do discounts affect profitability? What are the tipping points where discounts begin to negatively impact profit?
-

About Data

The dataset for this project is stored in the **shopcart_analytics** database across four main tables: **orders**, **customers**, **products**, and **shipping_address**.

- The **orders** table contains transactional details such as Order ID, Order Date, Ship Date, Ship Mode, Quantity, Sales, Discount, and Profit.
- The **customers** table includes information about each customer, including Customer ID, First name, Last Name, and the customer segment.
- The **products** table contains details about each product, such as Product ID, Product Name, Category, and Sub-Category.
- The **shipping_address** table contains location details, such as Postal Code, City, State, Country, and Region.

These tables are joined to create a single dataset for analyzing sales and profit trends in the United States between January 2022 and December 2025.

1) Importing Libraries

```
In [1]: from sqlalchemy import create_engine
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from matplotlib.patches import Patch
import os
```

```
In [2]: pd.set_option("display.max_columns",None)
```

```
In [3]: # Colours
orange = "#F59E0B"
blue = '#2196F3'
green = '#006400'
title_text = "#222021"
grid_color = '#BEBDB8'

colors = ["#344771", "#b3cde0", "#03396c", "#6497b1", "#011f4b", "#7b8dd2",]
```

2) Loading Data from Database & Basic Inspection

```
In [4]: # Logging into Database using credentials

host = "localhost"
password = os.getenv("MYSQL_PASS")
username = "root"
database = "ShopCart_Analytics"
port = "3306"

engine = create_engine(f"mysql+pymysql://{username}:{password}@{host}:{port}/{database}")
```

```
In [5]: query = """
        SELECT
            o.`Row ID`,
            o.`Order ID`,
            o.`Order Date`,
```

```
o.`Ship Date`,
o.`Ship Mode`,
c.`Customer ID`,
CONCAT(c.`First Name`, ' ', c.`Last Name`) AS `Customer Name`,
c.`Segment`,
a.`Postal Code`,
a.`City`,
a.`State`,
a.`Country`,
a.`Region`,
p.`Product ID`,
p.`Product Name`,
p.`Category`,
p.`Sub-Category`,
o.`Sales`,
o.`Quantity`,
o.`Discount`,
o.`Profit`
FROM shopcart_analytics.orders o
INNER JOIN shopcart_analytics.customers c
    ON o.`Customer ID` = c.`Customer ID`
INNER JOIN shopcart_analytics.products p
    ON o.`Product ID` = p.`Product ID`
INNER JOIN shopcart_analytics.shipping_address a
    ON o.`Address ID` = a.`Address ID`

WHERE
    a.`Country` = 'United States'
    AND o.`Order Date` BETWEEN '2022-01-01' AND '2025-12-31'
ORDER BY
    o.`Order Date`;

"""

# Loading Data
df = pd.read_sql(query, engine)
```

```
In [6]: df.head(3)
```

Out[6]:

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Postal Code	City	State	Country	Region
0	7981	CA-2022-103800	2022-01-03	2022-01-07	Standard Class	DP-13000	Darren Powers	Consumer	77095	Houston	Texas	United States	Central
1	740	CA-2022-112326	2022-01-04	2022-01-08	Standard Class	PO-19195	Phillina Ober	Home Office	60540	Naperville	Illinois	United States	Central
2	741	CA-2022-112326	2022-01-04	2022-01-08	Standard Class	PO-19195	Phillina Ober	Home Office	60540	Naperville	Illinois	United States	Central



```
In [7]: # Data Shape
print(f"Number of rows: {df.shape[0]}")
print(f"Number of columns: {df.shape[1]}")
```

Number of rows: 9994.

Number of columns: 21.

```
In [8]: # Basic Data Summary
df_info = pd.DataFrame({
    "dtype": df.dtypes,
    "null_count": df.isna().sum(),
    "null_prct": df.isna().mean() * 100,
    "unique_count": df.nunique()})

df_info
```

Out[8]:

	dtype	null_count	null_prct	unique_count
Row ID	int64	0	0.0	9994
Order ID	object	0	0.0	5009
Order Date	object	0	0.0	1237
Ship Date	object	0	0.0	1334
Ship Mode	object	0	0.0	4
Customer ID	object	0	0.0	793
Customer Name	object	0	0.0	793
Segment	object	0	0.0	3
Postal Code	int64	0	0.0	631
City	object	0	0.0	530
State	object	0	0.0	49
Country	object	0	0.0	1
Region	object	0	0.0	4
Product ID	object	0	0.0	1862
Product Name	object	0	0.0	1818
Category	object	0	0.0	3
Sub-Category	object	0	0.0	17
Sales	float64	0	0.0	5825
Quantity	int64	0	0.0	14
Discount	float64	0	0.0	12
Profit	float64	0	0.0	7287

```
In [9]: # Numeric Columns Summary
df.describe()
```

```
Out[9]:
```

	Row ID	Postal Code	Sales	Quantity	Discount	Profit
count	9994.000000	9994.000000	9994.000000	9994.000000	9994.000000	9994.000000
mean	4997.500000	55190.379428	229.858001	3.789574	0.156203	28.656896
std	2885.163629	32063.693350	623.245101	2.225110	0.206452	234.260108
min	1.000000	1040.000000	0.444000	1.000000	0.000000	-6599.978000
25%	2499.250000	23223.000000	17.280000	2.000000	0.000000	1.728750
50%	4997.500000	56430.500000	54.490000	3.000000	0.200000	8.666500
75%	7495.750000	90008.000000	209.940000	5.000000	0.200000	29.364000
max	9994.000000	99301.000000	22638.480000	14.000000	0.800000	8399.976000

```
In [10]: # Checking for Duplicates
df.duplicated().sum()
```

```
Out[10]: np.int64(0)
```

Insights

- No missing values detected in the dataset.
- No duplicate values detected in the dataset.

3) Data Cleaning and Pre-processing

```
In [11]: # 1) Standardizing column names (in snake case)
df.columns = df.columns.str.strip().str.replace(" ", "_").str.replace("-", "_").str.lower()
df.columns
```

```
Out[11]: Index(['row_id', 'order_id', 'order_date', 'ship_date', 'ship_mode',  
              'customer_id', 'customer_name', 'segment', 'postal_code', 'city',  
              'state', 'country', 'region', 'product_id', 'product_name', 'category',  
              'sub_category', 'sales', 'quantity', 'discount', 'profit'],  
             dtype='object')
```

```
In [12]: # 2) Parsing date columns  
df["order_date"] = pd.to_datetime(df["order_date"])  
df["ship_date"] = pd.to_datetime(df["ship_date"])
```

```
In [13]: # 3) Dropping country column (it contain only "USA.")  
df.drop(columns="country", inplace=True)
```

4) Feature Engineering

4.1) Creating new columns:

```
In [14]: # a) 'days_to_ship' - Shipping time in days  
df["days_to_ship"] = (df["ship_date"] - df["order_date"]).dt.days
```

```
In [15]: # b) 'profit_margin_pct' - Profit margin percentage  
df["profit_margin_pct"] = df["profit"] / df["sales"] * 100
```

```
In [16]: # c) 'revenue_per_unit' - revenue generated per unit sold  
df["revenue_per_unit"] = (df["sales"] / df["quantity"]).round(2)
```

```
In [17]: # d) 'discount_bucket' - categorizing discount into discount groups  
bins     = [-0.1, 0.001, 0.10, 0.20, 0.30, 0.40, 0.50, 1.0]  
labels   = ['No Discount', '1-10%', '11-20%', '21-30%', '31-40%', '41-50%', '>51%']  
df["discount_bucket"] = pd.cut(df["discount"], bins=bins, labels=labels)
```

5) Executive KPI Summary

In [18]: # ShopCart - Executive Summary

```
total_sales = round(df["sales"].sum())
total_profit = round(df["profit"].sum())
overall_margin = round(total_profit/ total_sales * 100,2)
total_orders = df["order_id"].nunique()
total_customers = df["customer_id"].nunique()
avg_ship_days = df["days_to_ship"].mean().round(2)
loss_making_orders = (df["profit"] < 0).sum()
loss_making_prct = ((df["profit"] < 0).mean() * 100).round(2)
avg_discount = df["discount"].mean().round(2) * 100

print("-" * 50)
print("      ShopCart - Executive Summary (2022-2025)")
print("-" * 50)
print(f"      Total Revenue:          ${total_sales}")
print(f"      Total Profit:           ${total_profit}")
print(f"      Overall Profit Margin:  {overall_margin} %")
print(f"      Average Discount:       {avg_discount} %")
print(f"      Loss-making orders:     {loss_making_orders} ({loss_making_prct} %)")
print(f"      Total Orders:           {total_orders}")
print(f"      Total Customers:        {total_customers}")
print(f"      Average Shipping Days:  {avg_ship_days}")
```

ShopCart - Executive Summary (2022-2025)

Total Revenue:	\$2297201
Total Profit:	\$286397
Overall Profit Margin:	12.47 %
Average Discount:	16.0 %
Loss-making orders:	1871 (18.72 %)
Total Orders:	5009
Total Customers:	793
Average Shipping Days:	3.96

6) Sales and Profit Trend over Time

```
In [19]: # 1) Yearly Sales and Profit Trend
yearly_summary = df.groupby(df["order_date"].dt.year).agg(
    Sales = ("sales", lambda x: round(x.sum())),
    Profit = ("profit", lambda x: round(x.sum())),
    Orders = ("order_id", "nunique")).reset_index()

yearly_summary["Profit Margin (%)"] = round(yearly_summary["Profit"] / yearly_summary["Sales"] * 100, 2)
yearly_summary["Sales YoY Change (%)"] = round(yearly_summary["Sales"].pct_change() * 100, 2)
yearly_summary.rename(columns={"order_date": "Year"}, inplace=True)

# Bar Graph
fig, axes = plt.subplots(figsize=(14, 5))
fig.suptitle("Annual Sales vs Profit Trend (2022-2025)", fontsize=16, fontweight="bold", color=title_text)

bars = axes.bar(yearly_summary["Year"].astype(str), yearly_summary["Sales"] / 1e3, label='Sales',
                color="#03396C", alpha=0.6, width=0.45, linewidth=2, edgecolor="#03396C")
axes.set_ylabel("Total Sales (in $K)", fontsize=11, fontweight="bold", color = title_text)

for bar in bars:
    h = bar.get_height() + 10
    x_pos = bar.get_x() + bar.get_width() / 2
    axes.text(x_pos, h, f"${bar.get_height():.0f}K", ha='center', va='bottom', fontsize=12)

ax2 = axes.twinx()
ax2.plot(yearly_summary["Year"].astype(str), yearly_summary["Profit"] / 1e3, label = "Profit",
        color= orange, marker='o', linewidth=3, markersize=9, markerfacecolor="#FFD27F",
        markeredgewidth=1.5)
ax2.set_ylabel("Total Profit (in $K)", fontsize=11, fontweight="bold", color = title_text)

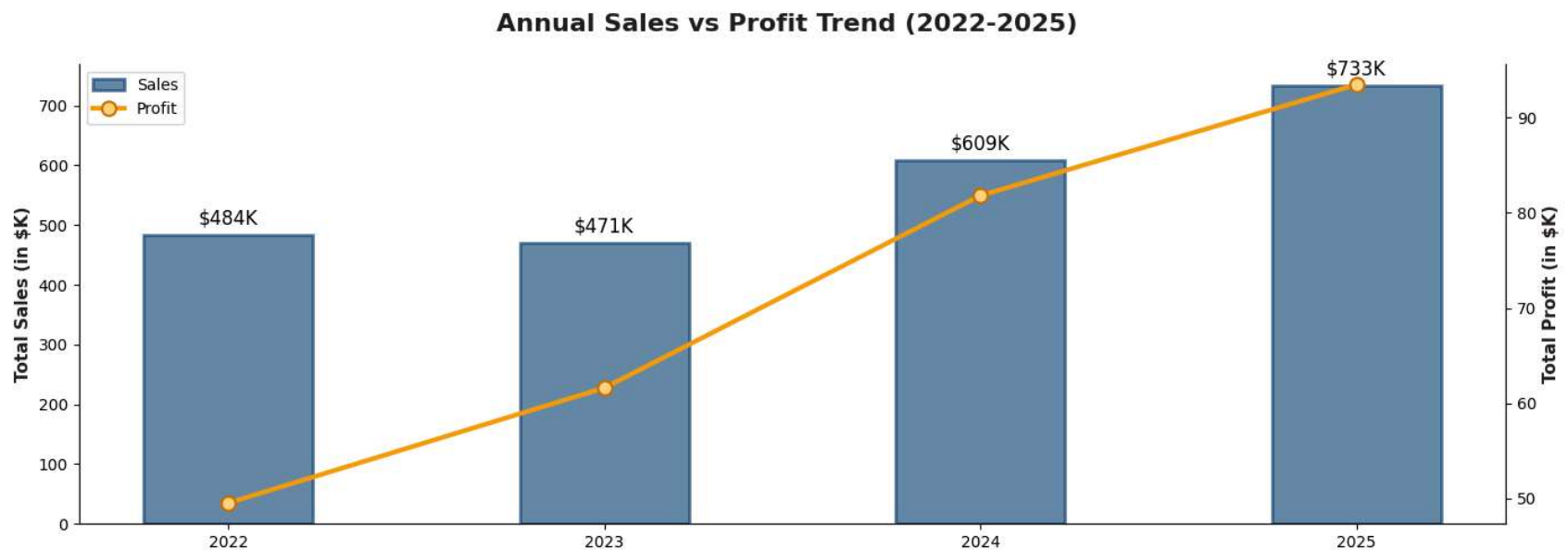
handles1, labels1 = axes.get_legend_handles_labels()
handles2, labels2 = ax2.get_legend_handles_labels()

axes.legend(handles1 + handles2, labels1 + labels2, loc='upper left', fontsize=10)
axes.spines["top"].set_visible(False)
```

```
ax2.spines["top"].set_visible(False)

plt.tight_layout()
plt.show()

print("\nYear Over Year Summary Table: ")
print(yearly_summary.to_string(index=False))
```



Year Over Year Summary Table:

Year	Sales	Profit	Orders	Profit Margin (%)	Sales YoY Change (%)
2022	484247	49544	969	10.23	NaN
2023	470533	61619	1038	13.10	-2.83
2024	609206	81795	1315	13.43	29.47
2025	733215	93439	1687	12.74	20.36

```
In [20]: # 2) Monthly Sales and Profit Trend
monthly = df.groupby(df["order_date"].dt.to_period("M")).agg(
    sales = ("sales", "sum"),
    profit = ("profit", "sum")).reset_index()
monthly["order_date"] = monthly["order_date"].dt.to_timestamp()

colors_rg = ['#4CAF50' if x>=0 else '#F44336' for x in monthly['profit']]
```

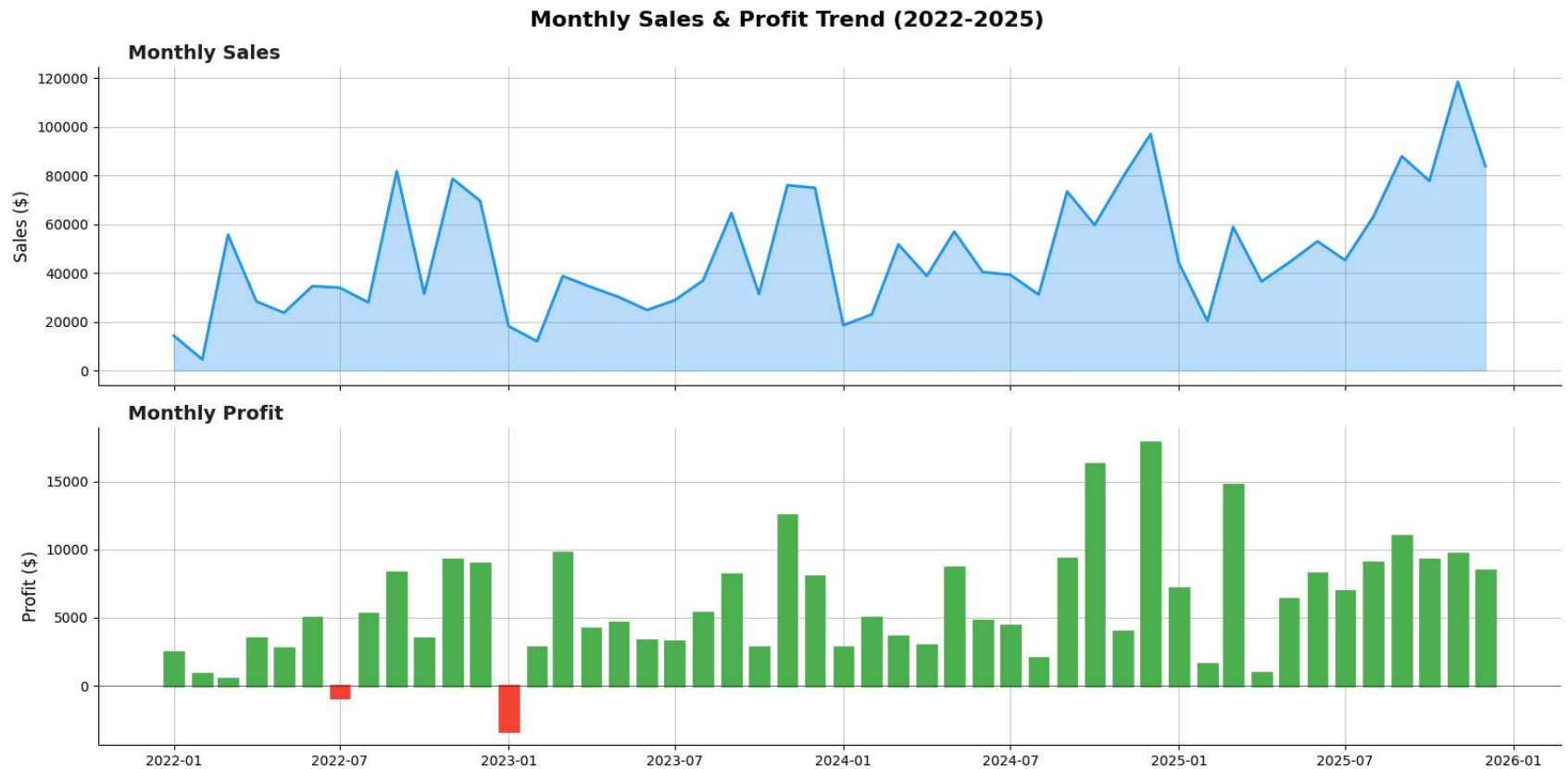
```
fig, axes = plt.subplots(2, 1, figsize = (16, 8), sharex=True)
fig.suptitle("Monthly Sales & Profit Trend (2022-2025)", fontsize= 16, fontweight="bold", color="black")

axes[0].plot(monthly["order_date"], monthly["sales"], linewidth=2, color= blue)
axes[0].fill_between(monthly["order_date"], monthly["sales"], alpha = 0.3, color = blue)
axes[0].set_title("Monthly Sales", fontsize=14, fontweight="bold", color=title_text, loc="left", x=0.02)
axes[0].set_ylabel("Sales ($) ", fontsize=12)

axes[1].bar(monthly["order_date"], monthly["profit"], width=20, color=colors_rg, linewidth=2, edgecolor=colo
axes[1].axhline(0, color = title_text, linewidth=0.4)
axes[1].set_title("Monthly Profit", fontsize=14, fontweight="bold", color=title_text, loc="left", x=0.02)
axes[1].set_ylabel("Profit ($) ", fontsize=12)

for ax in axes:
    ax.spines["top"].set_visible(False)
    ax.spines["right"].set_visible(False)
    ax.grid(linewidth=0.6,color=grid_color)
    ax.set_axisbelow(True)

plt.tight_layout()
plt.show()
```



Insights: Sales and Profit Trend over Time

- Both sales and profit shows an overall positive growth trend from 2022 to 2025, indicating consistent business expansion over time.
- This trend also shows strong seasonal patterns, with sales and profits consistently peaking between September and December for each year. This suggests increased customer demand and higher purchasing activity during the year-end and festival seasons.
- A few month periods shows losses, particularly in July 2022 and January 2023. These negative profit periods may indicate operational inefficiencies, higher costs, or the impact of aggressive discounting strategies.

7) Category and Sub-Category Analysis

```
In [21]: # 1) Sales and Profit Performace by Category
cat_summary = df.groupby("category").agg(
    Sales = ("sales", lambda x: round(x.sum())),
    Profit= ("profit", lambda x: round(x.sum())),
    Orders = ("order_id", "count"),
    Avg_Discount = ("discount", "mean")).reset_index()

cat_summary["Margin %"] = round(cat_summary["Profit"] / cat_summary["Sales"] * 100, 2)
cat_summary["Sales %"] = (cat_summary["Sales"] / cat_summary["Sales"].sum() * 100).round(2)
cat_summary["Profit %"] = (cat_summary["Profit"] / cat_summary["Profit"].sum() * 100).round(2)
cat_summary.sort_values("Profit", ascending=False, inplace=True)

# Chart
fig, axes = plt.subplots(1, 3, figsize=(14, 5))
fig.suptitle('Performance by Category', fontsize=18, fontweight='bold', color = "black")

charts = [ ("Sales", "Total Sales", "Sales (in $K)", '${:,.0f}K', 1e3),
           ("Profit", "Total Profit", "Profit (in $K)", '${:,.0f}K', 1e3),
           ("Margin %", "Profit Margin %", "Margin (%)", '{:.1f}%', 1)]

for ax, (col, chart_title, ylabel, fmt, scale) in zip(axes, charts):
    bars = ax.bar(cat_summary["category"], cat_summary[col] / scale, color=colors)
    ax.set_title(chart_title, fontsize=14, fontweight='bold', color=title_text, loc="left", x=0.1)
    ax.set_ylabel(ylabel, fontsize=10, fontweight='bold')

    for bar in bars:
        h = bar.get_height()
        x_pos = bar.get_x() + bar.get_width()/2
        ax.text(x_pos, h, fmt.format(h),
                ha='center', va='bottom', fontsize=9)

    ax.spines["top"].set_visible(False)
    ax.spines["right"].set_visible(False)
    ax.spines["left"].set_color("black")
```

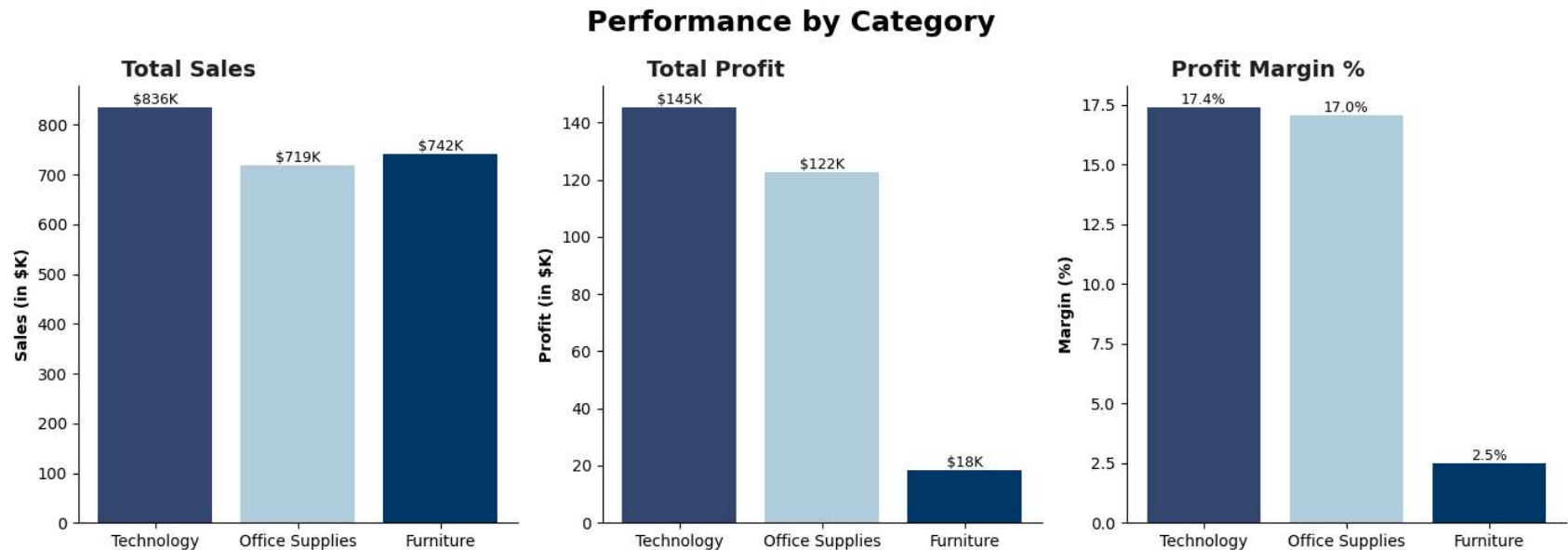
```

ax.spines["bottom"].set_color("black")
ax.grid(False)

plt.tight_layout()
plt.show()

print("\n Sales and Profit by Category: ")
print(cat_summary.to_string(index=False))

```



Sales and Profit by Category:

	category	Sales	Profit	Orders	Avg_Discount	Margin %	Sales %	Profit %
	Technology	836154	145455	1847	0.132323	17.40	36.4	50.79
	Office Supplies	719047	122491	6026	0.157285	17.04	31.3	42.77
	Furniture	742000	18451	2121	0.173923	2.49	32.3	6.44

Insights : Category wise Analysis

- **Technology is the strongest business driver**

- Technology is our top-performing category, It contributes 36% (846K USD) of total Sales and over 50% (145K USD) of total profit with a good profit margin of 17.5%.
- Despite accounting for only around one-third of sales, it generates more than half of the company's profit.
- It records the highest sales, highest profit, and highest contribution to overall profitability, making it the most valuable category for the business.
- **Office Supplies is also a strong business driver**
 - Office Supplies contributes around 31% (719K USD) of total sales and generates ~ 43% (122K USD) of total profit with a good margin of 17%.
 - Office Supplies also contributes the highest number of orders, indicating strong and consistent customer demand.
 - Office Supplies acts as a reliable revenue and profit generator for the business.
- **Furniture is the biggest concern**
 - Furniture contributes nearly 32% (742K USD) of total sales, which is comparable to Office Supplies, but generates only 6% (18K USD) of total profit.
 - It has the lowest profit margin 2.5% , making it significantly less profitable than the other categories.
 - Furniture has the highest average discount (17.3%), which is damaging profit.

- Technology and Office Supplies together contribute more than ****93% of total profit****, while Furniture contributes minimal despite high sales.

```
In [22]: # 2) Sub-category
sub_cat = (df.groupby(["category", "sub_category"]).agg(
    Sales=('sales', lambda x: round(x.sum())),
    Profit=('profit', lambda x: round(x.sum())),
    Avg_Discount_prct = ("discount", lambda x: round(x.mean() * 100,2))))
sub_cat["Margin_prct"] = sub_cat["Profit"] / sub_cat["Sales"] * 100
sub_cat = sub_cat.reset_index().sort_values("Sales")

category_colors = { "Furniture": colors[0], "Office Supplies": colors[1], "Technology": colors[2]}

bar_colors = sub_cat["category"].map(category_colors)
```

```

# Chart
fig, axes = plt.subplots(1, 2, figsize=(14, 6))
fig.suptitle('Sub-Category: Sales vs Profit', fontsize=18, fontweight='bold', color= "black")

charts = [ ("Sales", "Total Sales by Sub-Category", "Sales (in $K)", '${:,.1f}K', 1e3, (0, 360) ),
           ("Profit", "Total Profit by Sub-Category", "Profit (in $K)", '{:,.1f}%', 1e3, (-25, 60))]

for ax, (col, title_txt, xlabel, fmt, scale, xlim) in zip(axes, charts):
    bars = ax.barh(sub_cat["sub_category"], sub_cat[col] / scale, color=bar_colors, linewidth=1)
    for bar in bars:
        w = bar.get_width()
        y = bar.get_y() + bar.get_height() / 2
        ax.text(w, y, fmt.format(w),
                va='center', ha='left' if w >= 0 else 'right')

    ax.set_title(title_txt, fontsize=12, fontweight='bold', loc="left", color=title_text, x=0.05)
    ax.set_xlabel(xlabel, fontweight='bold')
    ax.set_xlim(xlim)
    ax.axvline(0, color='black', linewidth=0.8)

    ax.spines["top"].set_visible(False)
    ax.spines["right"].set_visible(False)
    ax.spines["left"].set_color("black")
    ax.spines["bottom"].set_color("black")
    ax.grid(False)

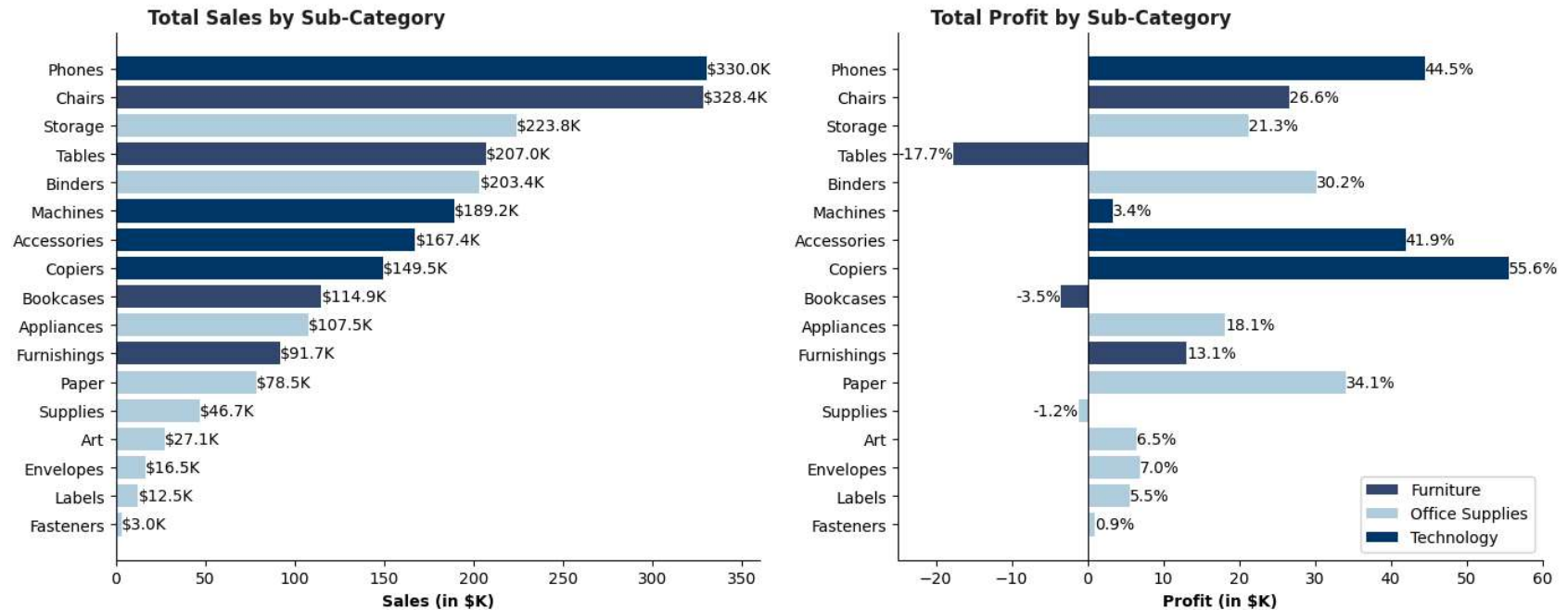
legends = [Patch(facecolor=color, label=category) for category, color in category_colors.items()]
axes[1].legend(handles=legends, loc='lower right', fontsize=10)

plt.tight_layout()
plt.show()

print("Three loss making sub-categories: ")
print(sub_cat.head(3).to_string(index=False))

```

Sub-Category: Sales vs Profit



Three loss making sub-categories:

	category	sub_category	Sales	Profit	Avg_Discount_prct	Margin_prct
Office Supplies	Fasteners	3024	950	8.20	31.415344	
Office Supplies	Labels	12486	5546	6.87	44.417748	
Office Supplies	Envelopes	16476	6964	8.03	42.267541	

Insights: Sub-category wise analysis

- The sub-category analysis reveals that profitability varies significantly even within categories.
- High performance sub-categories**
 - Copiers (Technology) and Phones (Technology) are the most profitable sub-categories, generating 55.6K USD and \$44.5K USD profit respectively.
 - Accessories (Technology) also performs strongly with a 25% margin, showing Technology is the most consistently profitable category.

- **High Profit margin sub-categories**

- Envelopes, Labels, Paper, and Fasteners show high profit margins (31%–44%) despite having very low sales volumes. All of these sub-categories belong to Office Supplies.

- **Loss Making Sub-Categories**

- Tables and Bookcases are both making losses, with profit margins of **-8.6% and -3.0%**. These both belong to the Furniture category.
- These sub-categories also have high average discounts (**26% and 21%**), which may be contributing to the losses.

-
- Profit is heavily concentrated in a few high-performing sub-categories, mainly within Technology and select Office Supplies products.
 - At the same time, a small number of Furniture sub-categories are significantly dragging overall profitability due to high discounts and negative margins.

8) Regional Performance Analysis

In [23]: *# 1) Regional Summary Table*

```
region_summary = df.groupby("region").agg(  
    Sales = ("sales", lambda x: round(x.sum())),  
    Profit = ("profit", lambda x: round(x.sum())),  
    Orders = ("order_id", "count"),  
    Customers = ("customer_id", "nunique"),  
    Discount_prct = ("discount", lambda x: round(x.mean() * 100, 2))).reset_index()  
  
region_summary["Margin (%)"] = (region_summary["Profit"] / region_summary["Sales"] * 100).round(1)  
region_summary = region_summary.sort_values("Profit", ascending=False)  
region_summary = region_summary.rename(columns={"region": "Region"})  
  
# Chart  
fig, axes = plt.subplots(1,3, figsize=(14,4))  
fig.suptitle('Sales and Profit by Region', fontsize=14, fontweight='bold')
```

```

charts = [ ("Sales", "Total Sales", "Sales (in $K)", 1e3, '${:,.0f}K'),
           ("Profit", "Total Profit", "Profit (in $K)", 1e3, '${:,.0f}K'),
           ("Margin (%)", "Profit Margin (%)", "Margin (%)", 1, '${:,.1f}%')]

for ax, (col, title_txt, ylabel, scale, fmt) in zip(axes, charts):
    bars = ax.bar(region_summary["Region"], region_summary[col]/ scale, color=colors)
    ax.set_title(title_txt, fontsize=12, fontweight='bold', loc="left", color=title_text, x=0.05)
    ax.set_ylabel(ylabel)

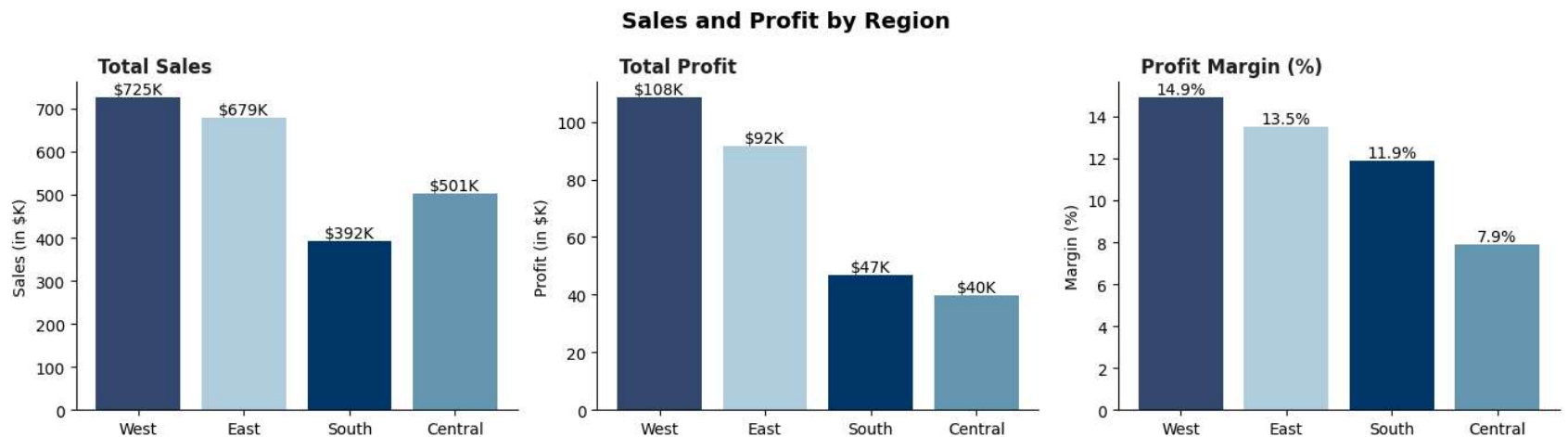
    for bar in bars:
        height = bar.get_height()
        x_pos= bar.get_x() + bar.get_width()/2
        ax.text(x_pos,height,fmt.format(height),
                ha='center',va='bottom')

    ax.spines["top"].set_visible(False)
    ax.spines["right"].set_visible(False)
    ax.spines["left"].set_color("black")
    ax.spines["bottom"].set_color("black")

plt.tight_layout()
plt.show()

print("\nSummary Statistics by Geographic Region: ")
print(region_summary.to_string(index=False))

```



Summary Statistics by Geographic Region:

Region	Sales	Profit	Orders	Customers	Discount_prct	Margin (%)
West	725458	108418	3203	686	10.93	14.9
East	678781	91523	2848	674	14.54	13.5
South	391722	46749	1620	512	14.73	11.9
Central	501240	39706	2323	629	24.04	7.9

Insights: Geographic Region Analysis

- **West is the strongest performing region**
 - It generates the highest sales (**725K USD**), highest profit (**108K USD**), and the highest profit margin (**14.9%**).
- **East and South shows moderate performance**
 - **East** is the second-best performing region after West.
 - **South** shows moderate performance in terms of both sales and profitability.
- **Central is the weakest region**
 - Despite generating **501K USD in sales**, it produces only **39.7K USD profit**, the lowest among all regions.
 - It has the **lowest margin (7.9%)**, indicating poor profitability conversion.
 - It also has the **highest average discount (24%)**, which is likely reducing profitability.

9) Customer Segment Analysis

In [24]: *# Customer Segement Analysis*

```

cust_seg = df.groupby("segment").agg(
    Sales = ("sales", lambda x: round(x.sum())),
    Profit = ("profit", "sum"),
    Orders = ("order_id", "count"),
    Customers = ("customer_id", "nunique"),
    Discount_prct = ("discount", lambda x: round(x.mean() * 100, 2)).reset_index()
cust_seg["Margin (%)"] = (cust_seg["Profit"] / cust_seg["Sales"] * 100).round(2)
cust_seg["Avg Order Value"] = (cust_seg["Sales"] / cust_seg["Orders"]).round(2)

```

```

# Chart
fig, axes = plt.subplots(1, 3, figsize=(14, 4))
fig.suptitle("Customer Segment Performance", fontsize=16, fontweight="bold")

axes[0].pie(cust_seg["Sales"], labels = cust_seg["segment"], autopct="%1.1f%", startangle=140,
            wedgeprops=dict(width=0.65, edgecolor='white'), colors=colors)
axes[0].set_title("Sales by Customer Segment", fontsize=12, fontweight="bold")

bars = axes[1].bar(cust_seg["segment"], cust_seg["Profit"] / 1e3, color=colors)
for bar in bars:
    height = bar.get_height()
    x_pos = bar.get_x() + bar.get_width() / 2
    axes[1].text(x_pos, height,
                 f'${height:.0f} K', ha='center', va='bottom',
                 fontsize=9, fontweight='bold')

axes[1].set_title("Total Profit by Segment", fontsize=12, fontweight="bold")
axes[1].set_ylabel("Profit (in $K)")
axes[1].spines["top"].set_visible(False)
axes[1].spines["right"].set_visible(False)

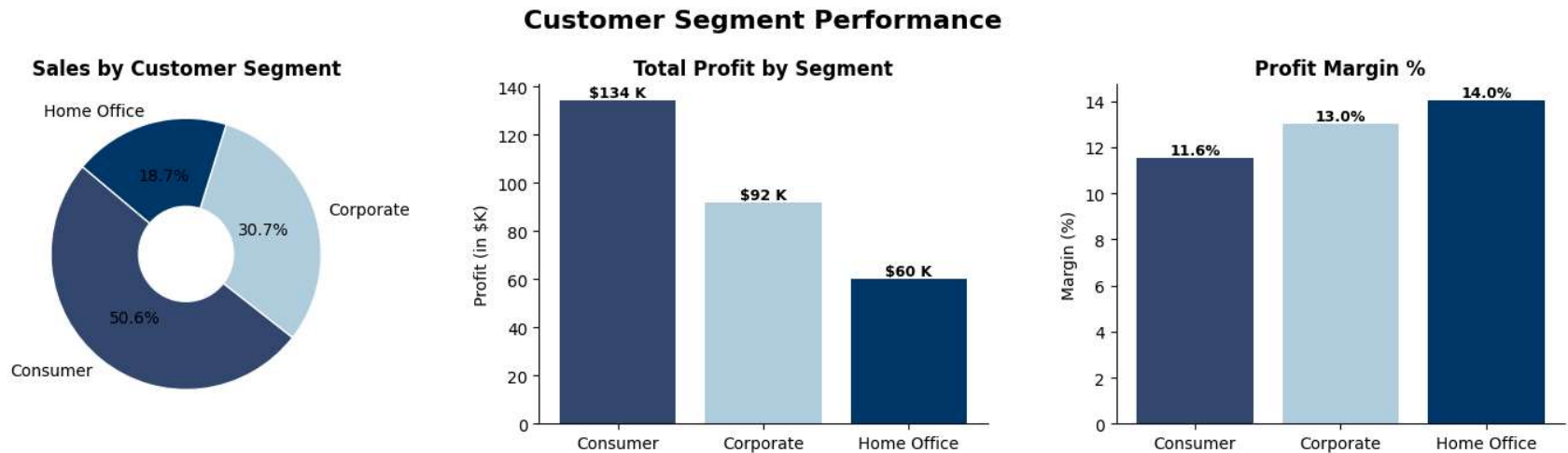
bars = axes[2].bar(cust_seg["segment"], cust_seg["Margin (%)"], color=colors)
for bar in bars:
    height = bar.get_height()
    x_pos = bar.get_x() + bar.get_width() / 2
    axes[2].text(x_pos, height,
                 f'{height:.1f}%', ha='center', va='bottom',
                 fontsize=9, fontweight='bold')

axes[2].set_title("Profit Margin %", fontsize=12, fontweight="bold")
axes[2].set_ylabel("Margin (%)")
axes[2].spines["top"].set_visible(False)
axes[2].spines["right"].set_visible(False)

plt.tight_layout()
plt.show()

print("\nCustomer Segment Performance: ")
print(cust_seg.to_string())

```



Customer Segment Performance:

	segment	Sales	Profit	Orders	Customers	Discount_prct	Margin (%)	Avg Order Value
0	Consumer	1161401	134119.2092	5191	409	15.81	11.55	223.73
1	Corporate	706146	91979.1340	3020	236	15.82	13.03	233.82
2	Home Office	429653	60298.6785	1783	148	14.71	14.03	240.97

Insight: Customer Segment Analysis

- **Consumer is the largest segment by scale**
 - Consumer segment generates highest **sales (1.16M USD)** and **profit (134K USD)** with the largest number of **orders (5,191)** and **customers (409)**.
 - However, it has the **lowest profit margin (11.55%)**, indicating lower efficiency despite high volume.
 - This segment is highly volume-driven but less efficient in profitability.
- **Corporate Segment is a moderate performer**
 - Corporate segment contributes **706K USD sales** and **92K USD profit**, making it the second-largest segment.
 - It has a better **margin (13.03%)** than Consumer, showing improved profitability efficiency.
 - It maintains a strong balance between volume and profitability, making it a stable segment.
- **Home Office Segment is the most efficient segment**

- Despite having the lowest sales (**429K USD**), it generates relatively strong **60K USD profit**.
- It has the **highest margin (14.03%)** and the **highest average order value (240.97)**.
- This indicates customers in this segment buy less frequently but with higher value and better profitability.

-
- All the customers segment have same average discount (~15%).
 - **Consumer drives volume but has the lowest efficiency**, making it a scale-heavy but margin-weak segment.
 - **Home Office is the most profitable per unit of sales**, showing strong value efficiency despite lower overall scale.
 - Overall, profitability improves as customer volume decreases, indicating that **high-volume segments are less efficient due to higher discounting or lower pricing power**.

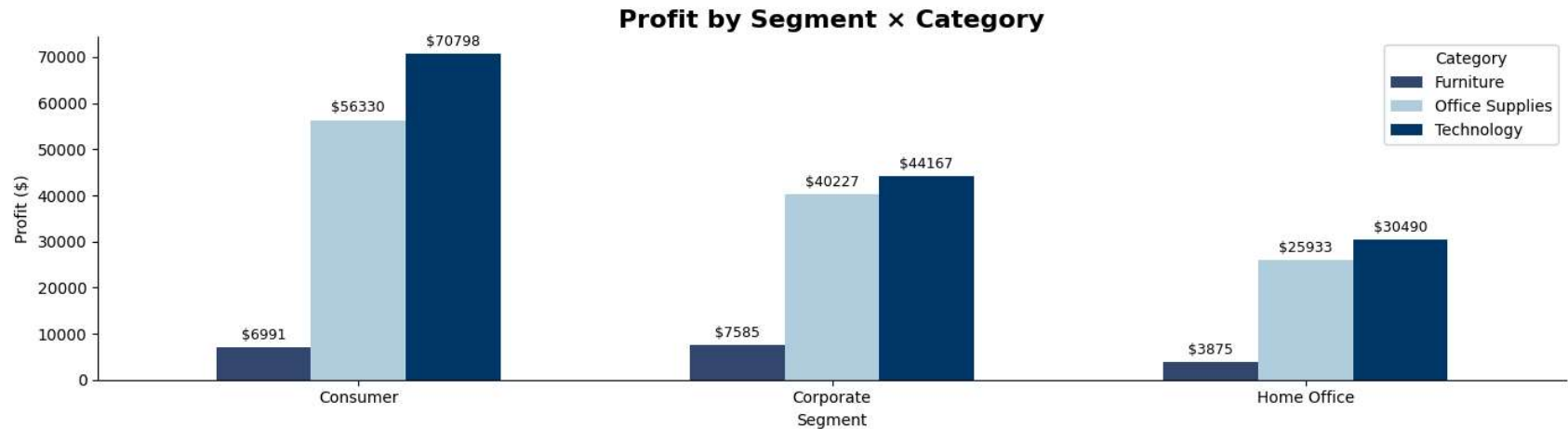
```
In [25]: # Profit by Segment x Category (Cross- Analysis)
seg_cat = (df.groupby(['segment', 'category'])['profit'].sum().unstack().fillna(0))

fig, ax = plt.subplots(figsize=(14, 4))
seg_cat.plot(kind='bar', ax=ax, color=colors, width=0.6)
ax.set_title('Profit by Segment x Category', fontsize=16, fontweight='bold')
ax.set_ylabel('Profit ($)'); ax.set_xlabel('Segment')
ax.legend(title='Category'); plt.xticks(rotation=0)

for container in ax.containers:
    ax.bar_label(container, fmt='%.0f', padding=3, fontsize=9)

ax.spines["top"].set_visible(False)
ax.spines["right"].set_visible(False)

plt.tight_layout()
plt.show()
```



Insights: Profit by Segment × Category

- **Technology is the most profitable category across all the customer segments**, generating the highest profit for each segments.
- **Consumer + Technology** is the strongest combination, contributing the highest overall profit (**70.8K**).
- **Office Supplies** also performs consistently well across all segments, making it a stable secondary profit driver.
- **Furniture is the weakest category across all the segments**, generating very low profit compared to Technology and Office Supplies.
- Even in the high-volume Consumer segment, Furniture contributes only **7K profit**, highlighting poor profitability efficiency.

10) Discount Impact Analysis

```
In [26]: # Discount impact Analysis
disc_bucket = df.groupby("discount_bucket", observed=True).agg(
    Orders = ("order_id", "count"),
    Sales = ("sales", lambda x: round(x.sum())),
    Profit = ("profit", lambda x: round(x.sum())),
    Avg_Profit = ("profit", lambda x: round(x.mean(), 2)).reset_index()
disc_bucket["Margin (%)"] = (disc_bucket["Profit"] / disc_bucket["Sales"] * 100).round(2)
```

```

# Fig
fig, axes = plt.subplots(2,2, figsize = (14,10))
fig.suptitle("Discount Impact on Profit Metrics", fontsize=16, fontweight="bold")

for ax, (col, title_, y_label, scale, fmt) in zip(axes.flatten(), [
    ("Orders", "Number of Orders", "Order Count", 1, '{:,.0f}'),
    ("Profit", "Total Profit", "Profit (in $K)", 1e3, '{:,.0f}K'),
    ("Avg_Profit", "Average Profit", "Profit", 1, '{:,.1f}'),
    ("Margin (%)", "Average Profit Margin", "Margin (%)", 1, '{:,.1f}%')]):

    colors_rg = ['#4CAF50' if x>=0 else '#F44336' for x in disc_bucket[col]]

    bars = ax.bar(disc_bucket["discount_bucket"], disc_bucket[col]/scale, color=colors_rg)
    ax.set_title(title_, fontsize=12, fontweight="bold", loc="left", x=0.1)
    ax.set_xlabel("Discount Bracket")
    ax.set_ylabel(y_label)
    for bar in bars:
        height = bar.get_height()
        x_pos = bar.get_x() + bar.get_width()/2
        ax.text(x_pos, height, fmt.format(height),
            ha='center', va='bottom' if height >= 0 else 'top')

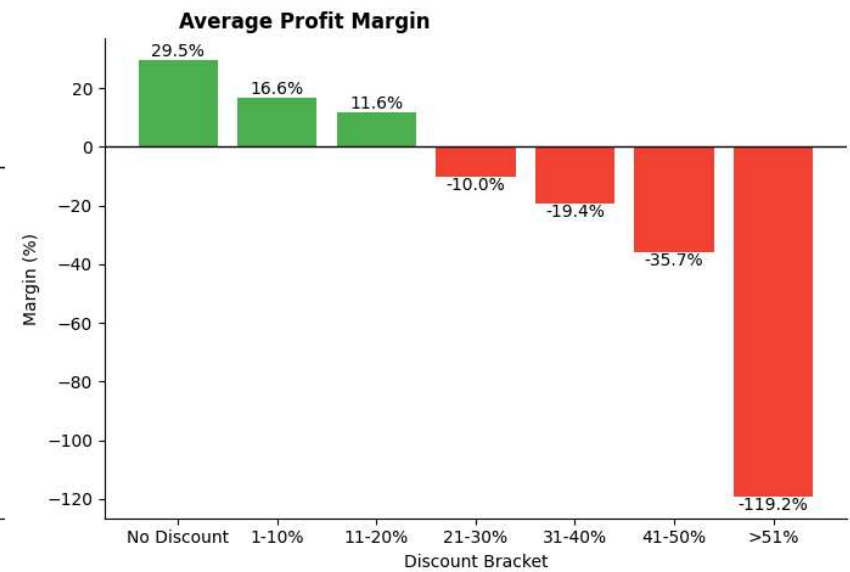
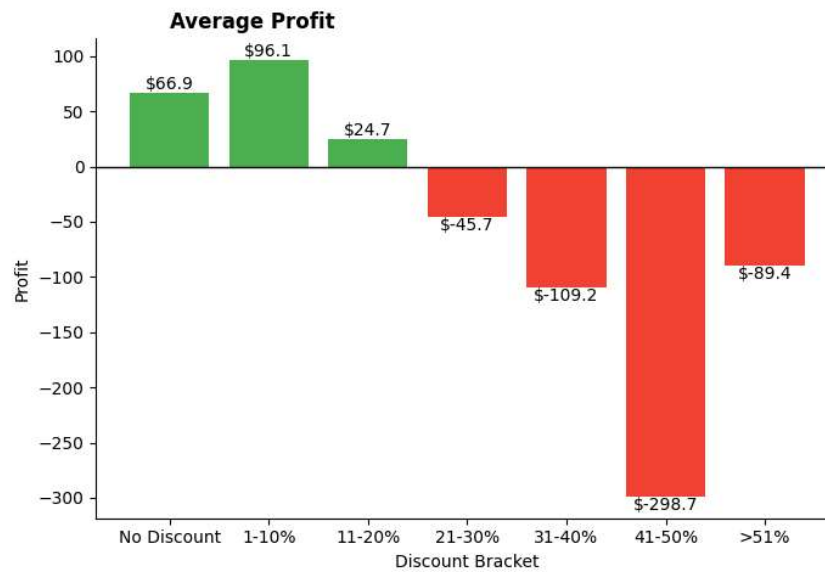
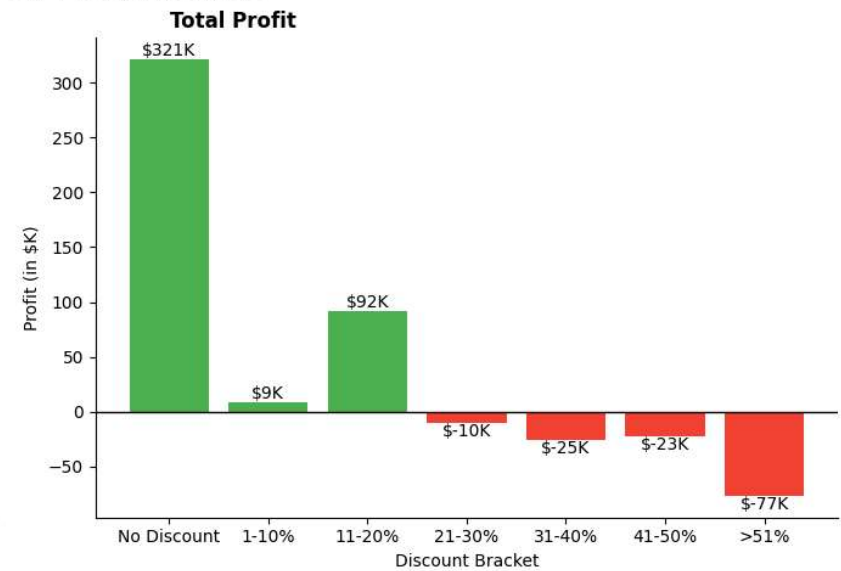
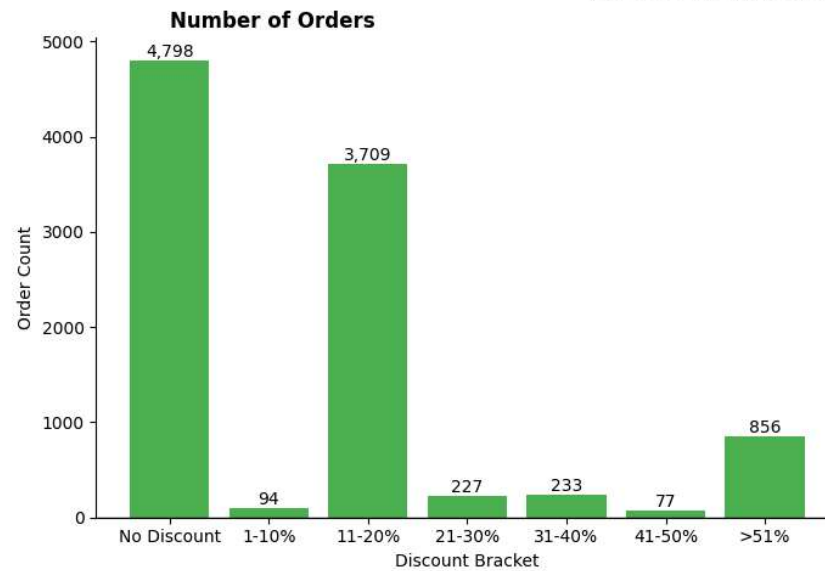
    ax.axhline(0, color='black', linewidth=1)
    ax.spines["top"].set_visible(False)
    ax.spines["right"].set_visible(False)

plt.tight_layout()
plt.show()

print("Discount Impact Analysis: ")
print(disc_bucket.to_string(index=False))

```

Discount Impact on Profit Metrics



Discount Impact Analysis:

discount_bucket	Orders	Sales	Profit	Avg_Profit	Margin (%)
No Discount	4798	1087908	320988	66.90	29.51
1-10%	94	54369	9029	96.06	16.61
11-20%	3709	792153	91756	24.74	11.58
21-30%	227	103227	-10369	-45.68	-10.04
31-40%	233	130911	-25448	-109.22	-19.44
41-50%	77	64404	-23000	-298.70	-35.71
>51%	856	64229	-76559	-89.44	-119.20

Insights: Discount Impact Analysis

- The discount analysis reveals a strong negative relationship between discount levels and profitability.
- The **No Discount** category is the most profitable segment, generating **~320K USD profit** with a **29.51% profit margin**, contributing the highest overall profitability.
- **Low discounts (1–10%)** remain efficient, maintaining a moderate **16.61% profit margin** while supporting sales growth.
- Profitability turns negative once discounts exceed **20%**, with all higher discount ranges consistently generating losses.
- The **>51% discount category** is the most damaging, showing severe declines in both profit margin and average profit per order.
- Overall, while heavy discounting may temporarily increase order volume, it significantly reduces margins and negatively impacts long-term profitability.

11) Product Level Analysis

```
In [27]: # Top 15 product by Analysis
top_products = df.groupby("product_name").agg(
    Sales = ("sales", lambda x: round(x.sum())),
    Profit = ("profit", lambda x: round(x.sum())),
    Quantity = ("quantity", "sum"),
    Discount_prct = ("discount", lambda x: round(x.mean() * 100, 2)).reset_index().sort_values("Sales", asce

top_products["product_name"] = top_products["product_name"].str[:25]
top_products["Margin (%)"] = (top_products["Profit"] / top_products["Sales"] * 100).round()
```

```
colors_rg = ['#4CAF50' if x>=0 else '#F44336' for x in top_products['Profit']]

# Fig
fig, axes = plt.subplots(1,2, figsize=(14,6))
fig.suptitle("Top 15 Products by Sales (2022-2025)", fontsize=16, fontweight="bold")

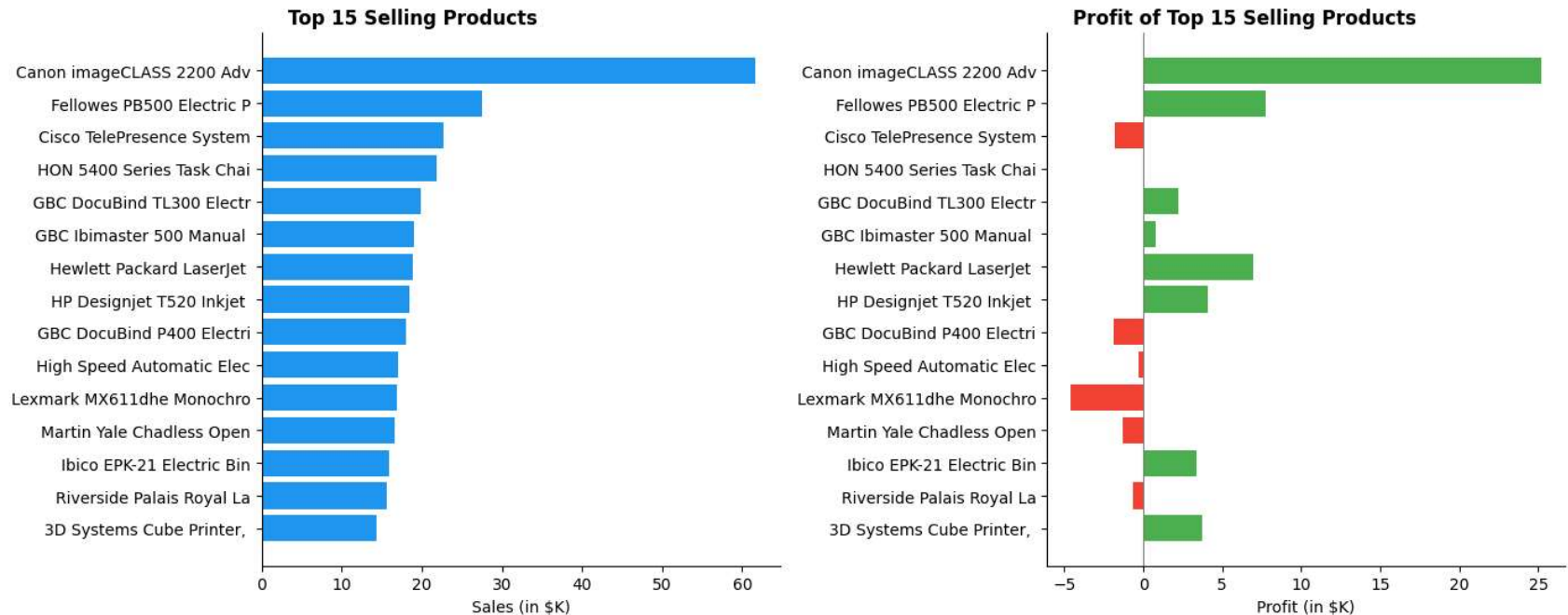
axes[0].barh(top_products["product_name"][::-1], top_products["Sales"][::-1] / 1e3, color=blue)
axes[0].set_title("Top 15 Selling Products", fontsize=12, fontweight="bold", loc="left", x=0.05)
axes[0].set_xlabel("Sales (in $K)")
axes[0].spines["top"].set_visible(False)
axes[0].spines["right"].set_visible(False)

axes[1].barh(top_products["product_name"][::-1], top_products["Profit"][::-1] / 1e3, color=colors_rg[::-1])
axes[1].set_title("Profit of Top 15 Selling Products", fontsize=12, fontweight="bold", loc="left", x=0.05)
axes[1].set_xlabel("Profit (in $K)")
axes[1].spines["top"].set_visible(False)
axes[1].spines["right"].set_visible(False)
axes[1].axvline(0, color='grey', linewidth=0.8 )

plt.tight_layout()
plt.show()

print("Top 10 Most Selling product Profit and Sales: ")
print(top_products.head(10).to_string(index=False))
```

Top 15 Products by Sales (2022-2025)



Top 10 Most Selling product Profit and Sales:

product_name	Sales	Profit	Quantity	Discount_prct	Margin (%)
Canon imageCLASS 2200 Adv	61600	25200	20	12.00	41.0
Fellowes PB500 Electric P	27453	7753	31	24.00	28.0
Cisco TelePresence System	22638	-1811	6	50.00	-8.0
HON 5400 Series Task Chair	21871	0	39	20.00	0.0
GBC DocuBind TL300 Electric	19823	2234	37	30.00	11.0
GBC Ibimaster 500 Manual	19024	761	48	52.22	4.0
Hewlett Packard LaserJet	18840	6984	38	20.00	37.0
HP Designjet T520 Inkjet	18375	4095	12	16.67	22.0
GBC DocuBind P400 Electric	17965	-1878	27	45.00	-10.0
High Speed Automatic Electric	17030	-262	11	6.67	-2.0

Insights: Top 15 Selling Products

- Top-selling products are not always the most profitable. While some products generate high sales and strong profits, several high-demand products contribute little or negative profit due to high costs or heavy discounting. Products

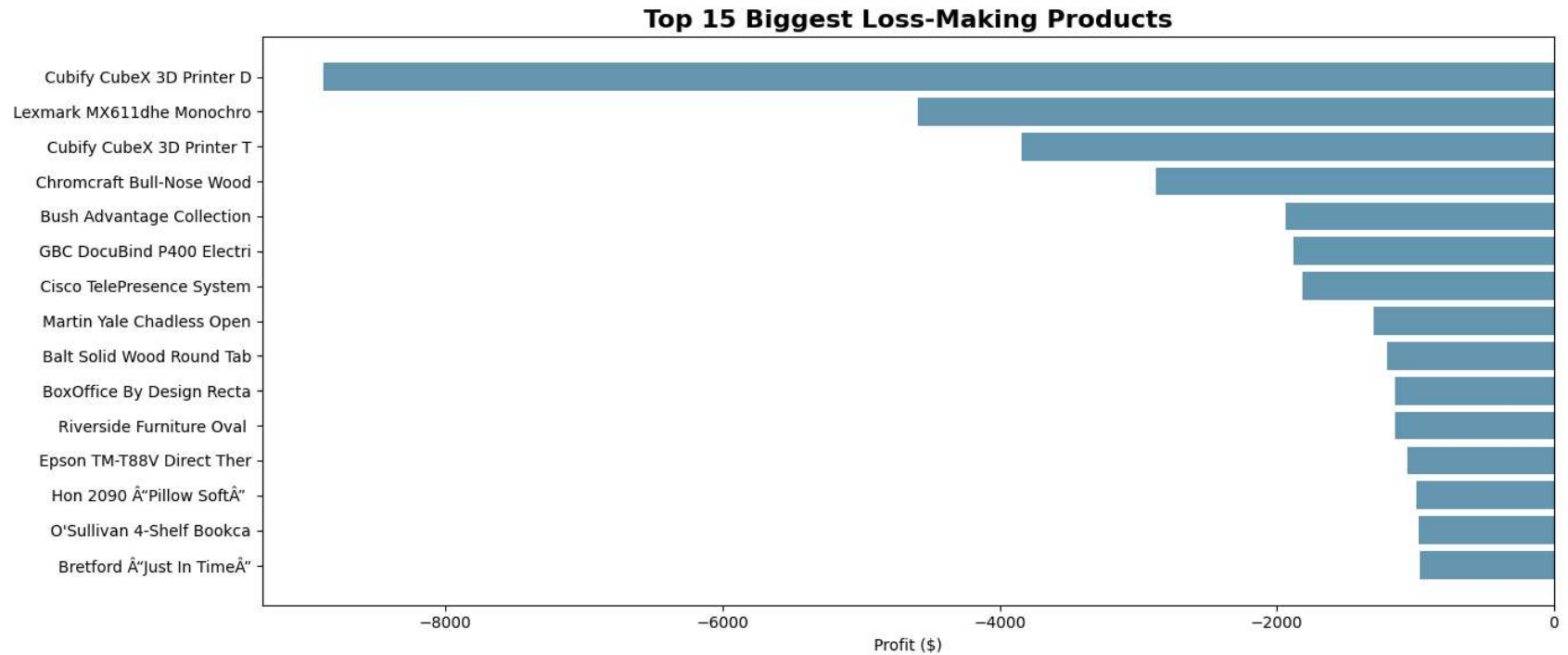
with strong sales and healthy profit margins should be promoted further, whereas consistently loss-making products should be reviewed for pricing, discount strategy, or potential discontinuation.

```
In [28]: # Top 15 worst products by Profit
worst_product = df.groupby("product_name").agg(
    Sales = ("sales", lambda x: round(x.sum())),
    Profit = ("profit", lambda x: round(x.sum())),
    Quantity = ("quantity", "sum"),
    Discount_prct = ("discount", lambda x: round(x.mean() * 100, 2)).reset_index().sort_values("Profit").head(15)

worst_product["Margin (%)"] = (worst_product["Profit"] / worst_product["Sales"] * 100).round()
worst_product["product_name"] = worst_product["product_name"].str[:25]

plt.figure(figsize=(14,6))
plt.title('Top 15 Biggest Loss-Making Products', fontsize=16, fontweight='bold')
plt.xlabel('Profit ($)')
plt.barh(worst_product["product_name"][::-1], worst_product["Profit"][::-1], color=colors[3])
plt.tight_layout()
plt.show()

print("Top 10 Loss making product")
print(worst_product.head(10).to_string(index=False))
```



Top 10 Loss making product

product_name	Sales	Profit	Quantity	Discount_prct	Margin (%)
Cubify CubeX 3D Printer D	11100	-8880	9	53.33	-80.0
Lexmark MX611dhe Monochro	16830	-4590	18	40.00	-27.0
Cubify CubeX 3D Printer T	8000	-3840	4	50.00	-48.0
Chromcraft Bull-Nose Wood	9918	-2876	27	28.00	-29.0
Bush Advantage Collection	9545	-1934	33	35.00	-20.0
GBC DocuBind P400 Electri	17965	-1878	27	45.00	-10.0
Cisco TelePresence System	22638	-1811	6	50.00	-8.0
Martin Yale Chadless Open	16656	-1299	22	10.00	-8.0
Balt Solid Wood Round Tab	6519	-1201	19	20.00	-18.0
BoxOffice By Design Recta	1706	-1148	15	48.33	-67.0

Insights: Top 15 worst performing products

The top loss-making products show a clear pattern of excessive discounting and weak profit margins. Most products with discounts above 40–50% generated significant losses despite moderate to high sales volumes. These products should be

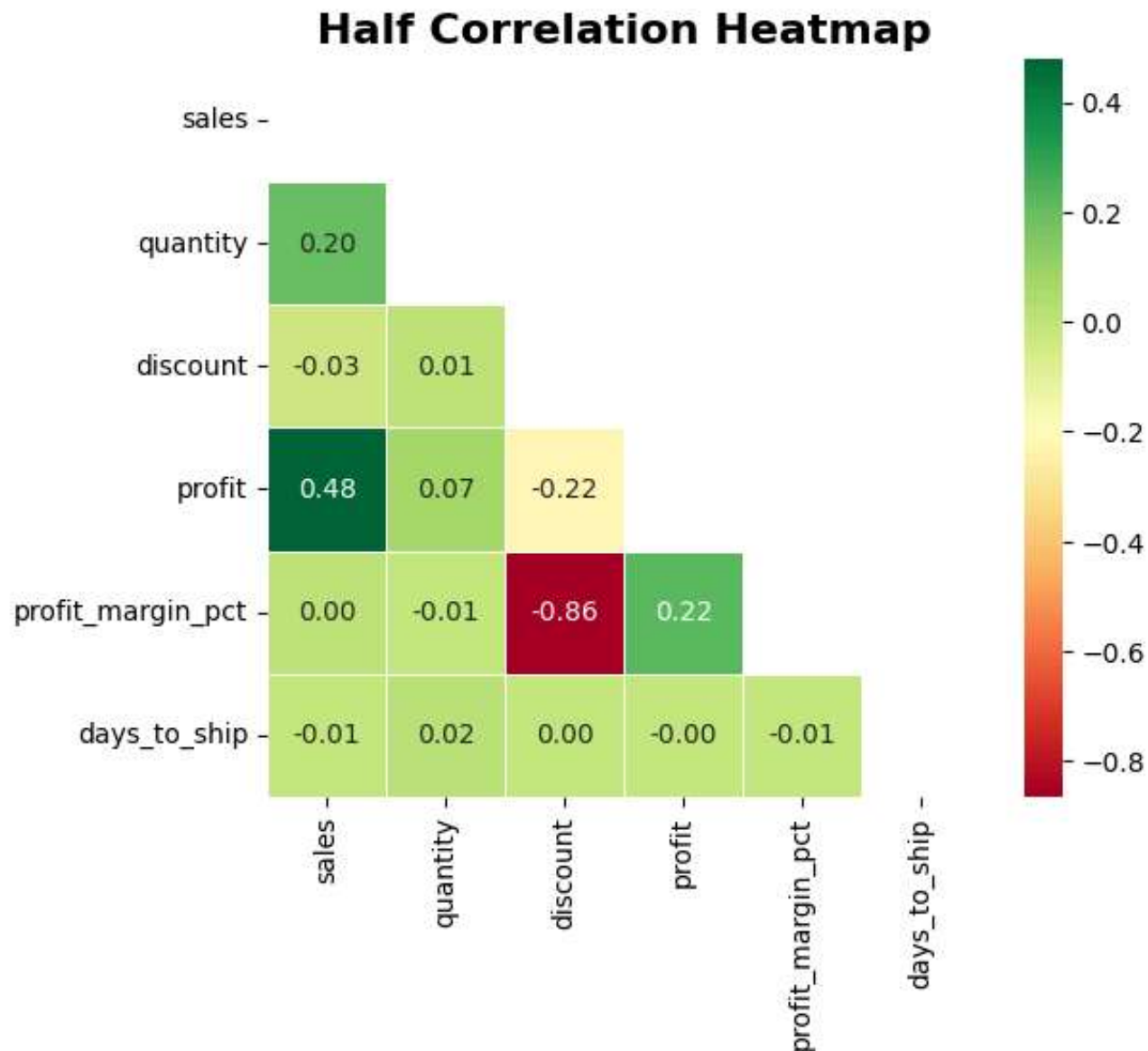
reviewed for pricing optimization, discount reduction, or possible discontinuation, while products with sustainable margins should receive greater promotional focus.

12) Correlation Heatmap

```
In [29]: columns = ["sales", "quantity", "discount", "profit", "profit_margin_pct", "days_to_ship"]
corr_matrix = df[columns].corr()

mask = np.triu(np.ones_like(corr_matrix, dtype=bool))

# Plot half heatmap
plt.figure(figsize=(6, 5))
sns.heatmap(corr_matrix, mask=mask, annot=True, cmap="RdYlGn", fmt=".2f", linewidth=0.5)
plt.title("Half Correlation Heatmap", fontsize=16, fontweight="bold")
plt.show()
plt.show()
```



Insights: Correlation Matrix

- **Discount and Profit Margin have strong negative correlation (-0.86).**
 - Discount shows a moderate negative correlation with profit (-0.22).

- This relationship indicates that increasing discount levels heavily reduces profit margins and overall profitability.
- **Sales and Profit have a moderate positive correlation (0.48)**, meaning that an increase in sales generally contributes to higher profits.
 - This relationship is not that strong, suggesting profitability also depends on other factors such as discounts.
- **Quantity sold has a very weak positive correlation with sales (0.20) and an almost negligible relationship with profit (0.07).**
 - This suggests that simply increasing the number of units sold does not guarantee higher profitability.
- **Profit margin percentage has weak positive correlation with profit (0.22),**
 - It indicates that products with higher margins do not always generate the highest total profit due to differences in sales volume.
- **Days to ship has almost no correlation with sales, profit, or profit margin.** -It shows that shipping time does not significantly influence financial performance in this dataset.

13) Key Findings and recommendataions

```
In [30]: print("-" * 50)
print("      ShopCart - Executive Summary (2022-2025)")
print("-" * 50)
print(f"      Total Revenue:          ${total_sales}")
print(f"      Total Profit:           ${total_profit}")
print(f"      Overall Profit Margin:   {overall_margin} %")
print(f"      Average Discount:       {avg_discount} %")
print(f"      Loss-making orders:      {loss_making_orders} ({loss_making_prct} %)")
print(f"      Total Orders:           {total_orders}")
print(f"      Total Customers:        {total_customers}")
print(f"      Average Shipping Days:   {avg_ship_days}")
```

ShopCart - Executive Summary (2022-2025)

Total Revenue:	\$2297201
Total Profit:	\$286397
Overall Profit Margin:	12.47 %
Average Discount:	16.0 %
Loss-making orders:	1871 (18.72 %)
Total Orders:	5009
Total Customers:	793
Average Shipping Days:	3.96

Recommendations

1) Invest more in the Technology Category

- Technology is the company's top-performing category. It contributes 36% (846K USD) of total sales and more than 50% (145K USD) of total profit while maintaining a strong profit margin of 17.5%.
- Technology delivers the highest sales and profit contribution among all categories, making it the most valuable category for the business.
- **Expanding the product range, improving stock availability, and prioritizing Technology products in marketing campaigns could further increase overall profitability.**
- **Increase investment in high-performing Technology sub-categories such as Copiers and Phones, which are leading contributors to profitability.**

2) Control Excessive Discounting

- The analysis clearly shows that profitability declines as discount levels increase.
- The **No Discount** segment is the most profitable segment, generating approximately **320K USD in profit** with a **29.5% profit margin**.
- Heavy discounting may temporarily increase sales volume, but it significantly reduces profit margins and overall profitability.
- On average discounts above 20% result in loss-making orders.

- **Discounting should be managed strategically rather than used aggressively to drive sales volume. Most discounts should remain within the 0–10% range.**
- **Introduce approval policies for discounts exceeding 20%. Controlling excessive discounting could significantly improve profit margins without heavily impacting revenue.**

3) Investigate the Furniture Category Thoroughly

- Furniture is one of the weakest-performing categories in terms of profitability. Despite generating strong sales, the category maintains a low average profit margin (~2.5%) and contains several loss-making sub-categories and products.
- Furniture also receives the highest average discount (~17.3%), which is significantly reducing profitability.
- Sub-categories such as Tables and Bookcases are major contributors to losses and receive some of the highest discount levels within the category.
- **The Furniture category requires deeper investigation to identify the root causes of continuous losses and understand why heavy discounting is necessary across several furniture sub-categories.**

4) Focus More on High-Margin Customer Segments

- The Consumer segment generates the highest sales volume (~50%) but generates comparatively weaker profit margins (~11.5%).
- In contrast, the Home Office segment generates stronger profit margins and higher average order values despite lower overall sales.
- This suggests that Home Office customers are more profitable on a per-order basis.
- **Increase marketing and retention efforts for Home Office and Corporate customers through targeted campaigns, personalized offers, and upselling strategies.**

5) Review the Central Region Performance

- Regional analysis shows that the East and West regions perform strongly in both sales and profitability.
- However, the Central region is underperforming and operates with significantly lower profit margins.
- The Central region also has relatively high discount levels (~24%), which are negatively impacting profitability.

- **A detailed investigation of the Central region is required to identify pricing issues, operational inefficiencies, and excessive discounting.**
- **Re-examine the discount strategy in the Central region — although sales are strong, profit margins remain significantly low.**

6) Prepare Earlier for Q4 Demand

- Our analysis shows a clear and recurring increase in sales during Q4 across all four years, making it one of the most important revenue-generating periods for the business.
- **The company should strengthen Q4 planning by building inventory earlier, optimizing logistics operations, and launching seasonal marketing campaigns ahead of peak demand.**

7) Monitor High-Sales but Low-Profit Products

- Several products generate strong revenue but consistently produce weak or negative profits, indicating that high sales do not always translate into strong business profitability.
- Not all top-selling products are highly profitable.
- **The company should regularly monitor product-level profitability alongside sales performance. Loss-making products should be reviewed for pricing strategy, discount dependency, shipping costs, and operational inefficiencies.**

8) Focus on High-Profit Categories

- Technology and several Office Supplies sub-categories generated the highest profits while maintaining stronger margins compared to Furniture products.
 - **ShopCart should prioritize high-margin categories and sub-categories when allocating marketing budgets, inventory investment, and promotional efforts to maximize overall profitability.**
-